

1. Texture Mapping Onto a Cube Face

Consider a cube with axis-aligned faces. For each face, we define:

- A point P_0 at the lower-left corner of the face
- A tangent vector T pointing across the face along the u direction
- A bitangent vector B pointing up the face along the v direction

For any hit point P on that face, the texture coordinates are obtained by projecting P onto the T and B axes:

$$u = \frac{(P - P_0) \cdot T}{\|T\|^2}, \quad v = \frac{(P - P_0) \cdot B}{\|B\|^2}.$$

If texture *repetition* (tiling) is desired, we apply:

$$u \leftarrow u - \lfloor u \rfloor, \quad v \leftarrow v - \lfloor v \rfloor.$$

This wraps (u, v) back into the unit square $[0, 1) \times [0, 1)$.

Finally, convert (u, v) to texture pixel coordinates for an image of width W and height H :

$$x = u(W - 1), \quad y = v(H - 1).$$

2. Defining the Frame for the +X Face

Assume the cube is centered at the origin with side length 2, so the $+X$ face is:

$$x = 1, \quad y \in [-1, 1], \quad z \in [-1, 1].$$

We choose:

$$P_0 = (1, -1, -1),$$

$$T = (0, 0, 2), \quad B = (0, 2, 0).$$

Thus:

$$\|T\|^2 = 4, \quad \|B\|^2 = 4.$$

3. Worked Examples

Example 1: Center of the Face

Let

$$P = (1, 0, 0).$$

Compute:

$$P - P_0 = (0, 1, 1).$$

Dot products:

$$(P - P_0) \cdot T = 2, \quad (P - P_0) \cdot B = 2.$$

Hence:

$$u = 2/4 = 0.5, \quad v = 2/4 = 0.5.$$

So the hit point maps to the center of the texture.

Example 2: Near the Top-Right

Let

$$P = (1, 0.8, 0.9).$$

Compute:

$$P - P_0 = (0, 1.8, 1.9).$$

Dot products:

$$(P - P_0) \cdot T = 3.8, \quad (P - P_0) \cdot B = 3.6.$$

Thus:

$$u = 3.8/4 = 0.95, \quad v = 3.6/4 = 0.9.$$

Example 3: A Large Wall (Texture Repeats)

Let

$$P = (1, 3, -5).$$

Compute:

$$P - P_0 = (0, 4, -4).$$

Dot products:

$$(P - P_0) \cdot T = -8, \quad (P - P_0) \cdot B = 8.$$

Thus raw coordinates:

$$u = -2, \quad v = 2.$$

Apply wrapping:

$$u \leftarrow u - \lfloor u \rfloor = -2 - (-2) = 0,$$

$$v \leftarrow v - \lfloor v \rfloor = 2 - 2 = 0.$$

The point lies exactly at the lower-left corner of a repeated tile of the texture.

4. Summary

For any hit point P on a cube face:

1. Select the appropriate (P_0, T, B) frame for that face.
2. Compute (u, v) using dot products.
3. Optionally wrap using modulo to tile the texture.
4. Convert (u, v) to pixel coordinates and sample the texture.

This method extends easily to rectangles of arbitrary orientation, and the only difference between cube faces is the choice of (P_0, T, B) .